Fraud and Detection Technologies – Tutorial Questions

Bachelor of Science in Cyber Security (3rd Year)

Section A: Foundations of Data Analysis

1. Types of Data Analysis

    1. Differentiate between quantitative and qualitative data analysis in fraud detection.

2. Explain the following quantitative analysis techniques with practical fraud-related examples:

    Sorting

    Categorization

    Relationship analysis

    Profiling

3. Why is qualitative analysis important in cyber fraud investigations?

4. Give two examples of:

    Contextual analysis

    Behavioural analysis

5. A bank suspects internal fraud. Which type of analysis would be more useful:

    Transaction trend analysis

    Employee behavioural analysis

    Justify your answer.

Section B: Performing Data Analysis

6. Gap Analysis

    1. Define gap analysis in fraud detection.

    2. How can gap analysis help identify missing financial transactions?

    3. A company's invoice numbers jump from INV1005 to INV1012. What possible fraud indicators may exist?

7. Duplicate and Fuzzy Duplicate Analysis

    1. Differentiate between:

    Duplicate analysis

    Fuzzy duplicate analysis

8. Explain how fuzzy duplicate testing can detect procurement fraud.

9. Give examples of data entries that may appear different but represent the same entity.

10. Orphan Transactions

1. What are orphan transactions?
2. Why are orphan transactions considered high-risk in financial systems?
3. Provide three examples of orphan transactions in:
    Payroll systems
    Inventory systems
    Banking systems

11. Timing Analysis
    1. Explain timing analysis and its importance in fraud detection.
    2. Why are transactions processed outside working hours considered suspicious?
    3. A user logs into a payroll system at 02:15 AM and authorizes payments. What investigative steps would you take?

12. Limit and Reasonableness Testing
    1. Define:
    Limit testing
    Reasonableness testing
    2. Explain how limit testing can detect credit card fraud.
    3. Why is reasonableness testing useful in payroll audits?

13. Trend and Outlier Analysis
    1. Explain the difference between trend analysis and outlier analysis.
    2. What types of fraud can be detected through trend analysis?
    3. A supplier's monthly payments suddenly increase by 300%. How would you investigate this anomaly?

14. Unwanted Match Analysis
    1. What is unwanted match analysis?
    2. Explain how matching employee addresses with supplier addresses may reveal fraud.
    3. Give three examples of suspicious matches in organizational databases.

15. Aging, Stratification, and Sampling
    1. Define:
        Aging analysis
        Stratification
        Sampling

2. Why is aging analysis important in debt recovery investigations?

3. Explain how sampling reduces audit workload while maintaining effectiveness.

Section C: Performing Data Profiling

16. Frequency and Intensity Tests

    1. What is a frequency test?

    2. Explain how intensity tests help identify suspicious user activity.

    3. A customer account performs 500 transactions within one hour. What conclusions may be drawn?

17. High-Risk Tests

    1. Define high-risk testing in fraud analytics.

    2. Identify five high-risk indicators in electronic payment systems.

    3. Why should transactions involving privileged users receive special attention?

18. Relative Size Tests

    1. Explain the purpose of relative size factor analysis.

    2. How can unusually large transactions indicate fraud?

    3. A junior employee authorizes payments larger than senior managers. What risks are present?

19. Digit Frequency Tests (Benford's Law)

    1. Explain Benford's Law and its application in fraud detection.

    2. Why do manipulated financial records often violate Benford's distribution?

    3. Give examples of datasets suitable for Benford analysis.

20. Concentration Tests (GEL)

    1. What is a concentration test?

    2. Explain how concentration analysis can identify procurement fraud.

    3. Why is vendor concentration considered risky?

Section D: Contextual and Behavioural Qualitative Analysis

21. Keyword and Content Analysis

    1. Explain the role of keyword searches in digital investigations.

    2. List five keywords investigators may search for during fraud investigations.

    3. How can image and file content analysis assist forensic investigations?

22. Behaviour Profiling

1. Define behavioural profiling in cyber investigations.
2. Explain how abnormal user behaviour may indicate insider threats.
3. Compare normal and suspicious behaviour in:

    Network access

    Email communication

    System usage

23. Online Presence and Social Media Analysis

1. Why is social media analysis important in fraud investigations?
2. Explain how investigators can ethically use online presence analysis.
3. Discuss legal and privacy concerns related to social media investigations.

24. Digital Forensics and Data Acquisition

1. Define digital forensic acquisition.
2. Differentiate between:

    Live acquisition

    Dead acquisition

3. Why is maintaining chain of custody important in digital investigations?
4. What are the risks of improper evidence handling?

Section E: OLAF Approach in Data Analysis

25. OLAF Methodology

1. Explain the OLAF approach:

    Object

    Locate

    Analyse

    Follow

2. How does the OLAF approach improve fraud investigations?
3. Apply the OLAF methodology to an online banking fraud scenario.

26. Miscarriage of Justice and Impurity of Findings

1. Define:

    Miscarriage of justice

    Impurity of findings

2. Explain how poor data analysis can lead to wrongful accusations.
3. What safeguards should investigators implement to maintain integrity?

Section F: Successful Implementation of Data Analysis

27. 3M Methodology
    1. Explain the "3M Methodology":
        Train the Captain
        Draw up the Map
        Train will Come
    2. Why is leadership important in fraud analytics implementation?
    3. Discuss challenges organizations face when implementing data analytics systems.
28. Critical Thinking and Logical Reasoning
    1. Why are critical thinking skills essential in fraud investigations?
    2. Differentiate between assumptions and evidence-based conclusions.
    3. Explain how logical reasoning supports digital forensic analysis.
29. Accounting and Commercial Knowledge
    1. Why should cyber fraud analysts understand accounting principles?
    2. Explain how financial statement analysis can reveal fraud.
    3. Discuss the relationship between accounting knowledge and cybercrime investigations.
30. Data Analysis Tools
    1. Differentiate between:
        Generic data analysis tools
        Specific fraud detection tools
    2. Identify three commonly used fraud analysis tools.
    3. What factors should organizations consider when selecting fraud analytics software?

Section G: Continuous Monitoring and Digital Surveillance

31. Monitoring Software
    1. Explain the role of monitoring software in fraud prevention.
    2. What are the advantages and disadvantages of continuous monitoring?
    3. How can real-time alerts reduce organizational losses?
32. Social Engineering
    1. Define social engineering.

2. Explain common social engineering techniques used by fraudsters.

3. How can organizations defend against phishing attacks?

33. Digital Surveillance Ethics

1. Discuss ethical concerns associated with digital surveillance.

2. How can organizations balance security and employee privacy?

3. Explain legal limitations of workplace monitoring.

Section H: Beyond Data Analysis and Digital Forensics

34. Findings Confirmation

1. Why is confirmation of findings important before accusing suspects?

2. Explain methods used to validate investigative findings.

3. What are the dangers of false positives in fraud detection?

35. Root Cause Analysis

1. Define root cause analysis.

2. Explain how identifying root causes prevents future fraud.

3. A company experiences repeated payroll fraud. Describe how you would perform root cause analysis.

36. Hypothesis Development

1. What is an investigative hypothesis?

2. Explain how hypotheses guide fraud investigations.

3. Develop a hypothesis for the following scenario:

"Multiple refunds are processed by the same cashier shortly before closing time."

37. Interview and Interrogation

1. Differentiate between:

Interview

Interrogation

2. Explain ethical guidelines investigators must follow during interrogations.

3. Why is evidence preparation important before interviewing suspects?

Practical / Case Study Questions

Case Study 1: Payroll Fraud

A company discovers that salary expenses increased by 40% over three months without hiring new employees.

Questions:

1. Which data analysis techniques would you apply?
2. How would you identify ghost employees?
3. What profiling tests would be useful?
4. Explain how digital forensic acquisition may support the investigation.

Case Study 2: Procurement Fraud

Several suppliers receive repeated payments just below approval limits.

Questions:

1. Which fraud indicators are present?
2. How would limit testing help?
3. Explain how Benford analysis could be applied.
4. What behavioural indicators may suggest collusion?

Case Study 3: Insider Threat

An employee accesses confidential customer records at unusual times and transfers files to external devices.

Questions:

1. What behavioural analysis techniques would apply?
2. What digital evidence should investigators collect?
3. Explain how continuous monitoring could have detected the incident earlier.
4. What legal and ethical issues must investigators consider?

Research / Discussion Questions

1. Discuss the impact of Artificial Intelligence in fraud detection.
2. Explain how machine learning improves anomaly detection.
3. Evaluate the effectiveness of continuous monitoring systems in modern organizations.
4. Discuss challenges facing digital forensic investigations in developing countries.
5. Debate: “Employee monitoring improves security more than it violates privacy.”